Homework

As in the Black-Scholes model, assume the stock price has dynamics

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t,$$

where $W = \{W_t\}_{t \leq T}$ is a Brownian motion under the physical measure $\mathbb{P}$. The interest rate is $r > 0$. Let $T$ be the maturity and $K$ the strike of a put option, and set $P^{BS}(t, x)$ as the price of the call given $S_t = x$. I.e.

$$P^{BS}(t, x) = \mathbb{E}^{\mathbb{Q}} \left[ e^{-r(T-t)} \left( K - S_T \right)^+ \middle| S_t = x \right]. \tag{1}$$

where $\mathbb{Q}$ is the risk neutral measure under which $S$ has drift $r$. The Black-Scholes formula states (you DO NOT have to prove this)

$$P^{BS}(t, x) = x(N(d_1(T-t, x)) - 2) + Ke^{-r(T-t)}(1 - N(d_2(T-t, x))),$$

where $N$ is the standard normal cdf and

$$\begin{aligned} d_1(\tau, x) &= \frac{1}{\sigma\sqrt{\tau}} \left( \log\left(\frac{x}{K}\right) + \left( r + \frac{1}{2}\sigma^2 \right)(\tau) \right), \\ d_2(\tau, x) &= d_1 - \sigma\sqrt{\tau}. \end{aligned}$$

Furthermore, with $\phi$ denoting the standard normal pdf we have

$$\begin{aligned} \delta(t, x) &= \partial_x P^{BS}(t, x) = N(d_1) - 1, \\ \gamma(t, x) &= \partial_{xx} P^{BS}(t, x) = \frac{\phi(d_1(T-t, x))}{x\sigma\sqrt{T-t}}, \\ \theta(t, x) &= \partial_t P^{BS}(t, x) = -\frac{\sigma}{2\sqrt{T-t}} x\phi(d_1(T-t, x)) \\ &\qquad + Kre^{-r(T-t)}(1 - N(d_2(T-t, x))). \end{aligned}$$

These are the "delta", "gamma" and "theta" respectively for the option.

At time $t$ we are short $M$ put options and long $M\delta(t, S_t)$ shares of $S$. Over the period $[t, t+\Delta]$ we hold the share position constant, writing $\lambda = \delta(t, S_t)$ to reinforce this fact. With this notation, the values of our portfolio at $t$ and $t + \Delta$ are

$$\begin{aligned} V_t &= M\left( \lambda S_t - P^{BS}(t, S_t) \right), \\ V_{t+\Delta} &= M\left( \lambda S_{t+\Delta} - P^{BS}(t+\Delta, S_{t+\Delta}) \right). \end{aligned}$$

1. With $z_t = \ln(S_t)$, identify the full, linearized, and second order loss operators over $[t, t+\Delta]$ as a function of the log return $x = X_{t+\Delta}$.
   (a) Make sure to fully evaluate the linearized loss operator - there is a cool answer!.
   (b) For the second order loss operator, only include the second derivative with respect to $x$: i.e. ignore the second order time derivative and second order time-space derivative.

2. Write a simulation which identifies the loss distribution for the portfolio using the full, linearized and second order (with the above djustments) loss operators. As in class, produce a histogram approximation of the probability density functions. How well do the approximations work?

   For parameters use $\mu = 0.16905$, $\sigma = 0.4907$, $r = 0.0011888$, $t = 0$, $T = .291667$, $\Delta = 10/252$ (ten day horizon), $S_0 = 152.51$, $K = 170$ and $M = 100$ options. Run $N = 100,000$ trials in your simulation.